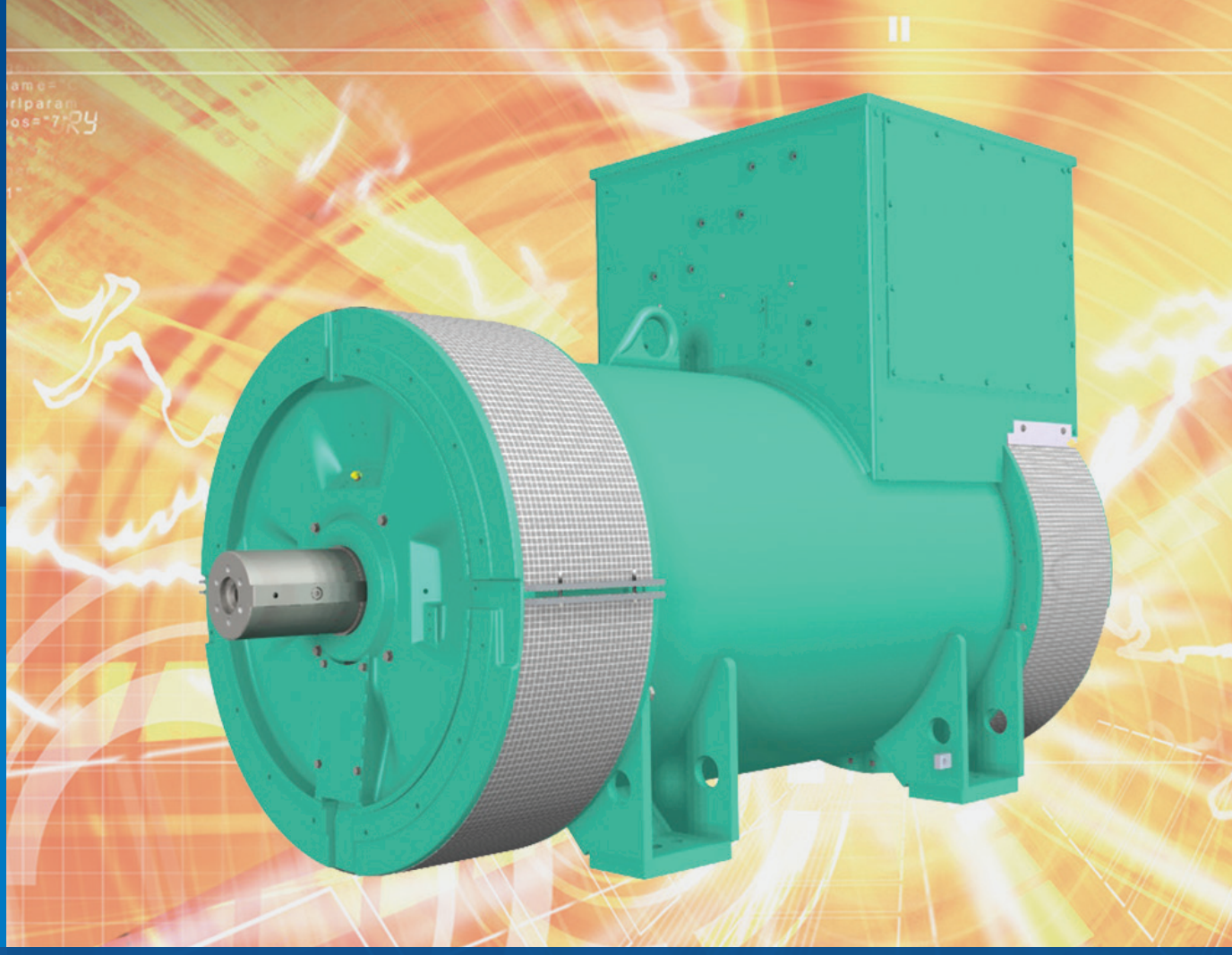


Low Voltage Alternators - 4 pole

LSA 52.3

1860 to 2500 kVA - 50 Hz / 2230 to 3000 kVA - 60 Hz
Electrical and mechanical data

Leroy-Somer™

EMERSON™
Industrial Automation

Low Voltage Alternators - 4 pole

LSA 52.3 - 1860 to 2500 kVA - 50 Hz / 2230 to 3000 kVA - 60 Hz

Specially adapted to applications

The LSA 52.3 alternator is designed to be suitable for typical generator applications, such as: backup, marine applications, rental, telecommunications, etc.

Compliant with international standards

The LSA 52.3 alternator conforms to the main international standards and regulations:
- IEC 60034, NEMA MG 1.32-33, ISO 8528-3, CSA / UL 1446 (UL 1004 on request), marine regulations, etc.

It can be integrated into a CE marked generator.
The LSA 52.3 is designed, manufactured and marketed in an ISO 9001 environment and ISO 14001.

Top of the range electrical performance

- Class H insulation.
- Standard 6-wire winding, 2/3 pitch, type no. 6S.
- Voltage range 50 Hz: 380V - 400V - 415V - 440 V.
- Voltage range 60 Hz: 380V - 416V - 440V - 480V.
- High efficiency and motor starting capacity.
- Other voltages are possible with optional adapted windings:
  - 50 Hz : 440 V (no. 7S), 500 V (no. 9S), 600 V (no. 22S or 23S), 690 V (no. 10S or 52S).
  - 60 Hz : 380 V and 416 V (no. 8S), 600 V (no. 9S).
- R 791 interference suppression conforming to standard EN 55011 group 1 class B standard for European zone (CE marking).

Excitation and regulation system suited to the application

The LSA 52.3 can be supplied with AREP + PMI or PMG excitation system, according to the alternator specification.
Standard excitation system is AREP with D 510C A.V.R.

Excitation system			Regulation options				
Volage regulator	AREP + PMI	PMG	Current transformer for paralleling	Mains paralleling	3-phase sensing	3-phase sensing for mains paralleling unbalanced	Remote voltage potentiometer
D510C	Std	Option	C.T.	included	included	contact factory	Option
R449	Option	Option	C.T.	R 726	R731	R734	Option

Protection system suited to the environment

- The LSA 52.3 is IP 23.
- Standard winding protection for clean environments with relative humidity ≤ 95 %, including indoor marine environments.

Options :
- Filters on air inlet : derating 5%.
- Filters on air inlet and air outlet (IP 44) : derating 8%.
- Winding protections for harsh environments and relative humidity greater than 95%.
- Space heaters.
- Protection or metering CTs.
- Thermal protection for stator windings &/or bearings (PT100).

Reinforced mechanical structure using finite element modelling

- Compact and rigid assembly to better withstand generator vibrations.
- Steel frame.
- Cast iron flanges and shields.
- Twin-bearing and single-bearing versions designed to be suitable for engines on the market.
- Half-key balancing.
- Regreasable bearings.
- Clockwise rotation in standard.

Accessible terminal box proportioned for optional equipment

- Easy access to the voltage regulator and to the connections.
- Possible inclusion of accessories for paralleling, protection and measurement.

Low Voltage Alternators - 4 pole

LSA 52.3 - 1860 to 2500 kVA - 50 Hz / 2230 to 3000 kVA - 60 Hz

General characteristics

Insulation class	H	Excitation system	AREP + PMI
Winding pitch	2/3 (n° 6S)	AVR type	D 510C
Number of wires	6	Voltage regulation (*)	± 0.5 %
Protection	IP 23	Short-circuit current	300% (3 IN) : 10s
Altitude	≤ 1000 m	Totale Harmonic distortion THD (**)	< 4 %
Overspeed	2250 min^{-1}	Waveform: NEMA = TIF (**)	< 50
Air flow	2.5 m^3/s (50 Hz) - 2.8 m^3/s (60 Hz)	Waveform: I.E.C. = THF (**)	< 2 %

(*) steady state (**) between phases

Ratings 50 Hz - 1500 R.P.M.

kVA / kW - P.F. = 0.8

Duty/T°C		Continuous duty/40°C				Continuous duty/40°C				Stand-by/40°C				Stand-by/27°C			
Class/T°K		H/125°K				F/105°K				H/150°K				H/163°K			
Phase		**3 ph.**				**3 ph.**				**3 ph.**				**3 ph.**			
Y		380V	400V	415V	440V	380V	400V	415V	440V	380V	400V	415V	440V	380V	400V	415V	440V
LSA 52.3 S5	kVA		**1860**		1692		**1700**		1547		**1975**		1800		**2050**		1870
	kW		*1488*		*1354*		*1360*		*1238*		*1580*		*1440*		*1640*		*1496*
LSA 52.3 S6	kVA		**2000**		1865		**1870**		1702		**2173**		1980		**2255**		2055
	kW		*1600*		*1492*		*1496*		*1361*		*1738*		*1584*		*1804*		*1644*
LSA 52.3 S7	kVA		**2200**		2002		**2006**		1825		**2332**		2122		**2420**		2205
	kW		*1760*		*1602*		*1604*		*1460*		*1865*		*1698*		*1936*		*1764*
LSA 52.3 L9	kVA		**2360**		2147		**2150**		1956		**2500**		2275		**2600**		2370
	kW		*1888*		*1717*		*1720*		*1565*		*2000*		*1820*		*2080*		*1896*
LSA 52.3 L12	kVA		**2500**		2275		**2280**		2075		**2650**		2385		**2750**		2505
	kW		*2000*		*1800*		*1824*		*1660*		*2120*		*1908*		*2200*		*2280*

Ratings 60 Hz - 1800 R.P.M.

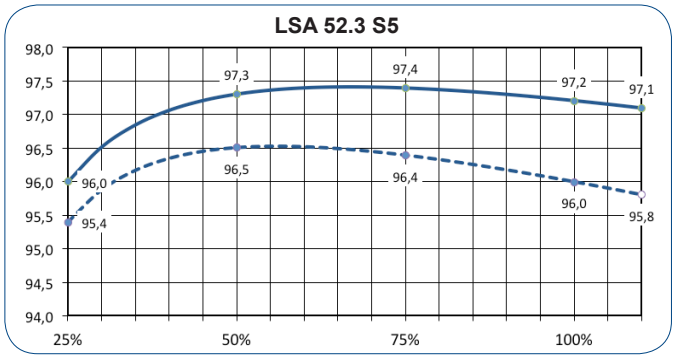
kVA / kW - P.F. = 0.8

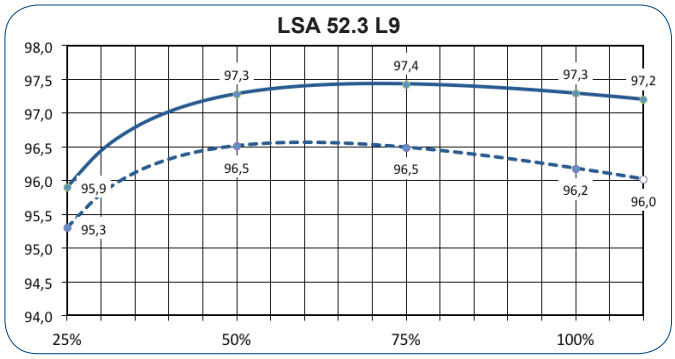
Duty/T°C		Continuous duty/40°C				Continuous duty/40°C				Stand-by/40°C				Stand-by/27°C			
Class/T°K		H/125°K				F/105°K				H/150°K				H/163°K			
Phase		**3 ph.**				**3 ph.**				**3 ph.**				**3 ph.**			
Y		380V	416V	440V	480V	380V	416V	440V	480V	380V	416V	440V	480V	380V	416V	440V	480V
LSA 52.3 S5	kVA	1860	1933	2042	**2230**	1700	1764	1864	**2035**	1975	2050	2166	**2365**	2050	2128	2250	**2455**
	kW	*1488*	*1546*	*1634*	*1784*	*1360*	*1411*	*1491*	*1628*	*1580*	*1640*	*1733*	*1892*	*1640*	*1703*	*1800*	*1964*
LSA 52.3 S6	kVA	2050	2133	2253	**2400**	1870	1946	2056	**2244**	2173	2261	2390	**2608**	2255	2346	2480	**2706**
	kW	*1640*	*1706*	*1803*	*1920*	*1496*	*1556*	*1644*	*1795*	*1738*	*1810*	*1911*	*2086*	*1804*	*1877*	*1983*	*2164*
LSA 52.3 S7	kVA	2200	2290	2418	**2640**	2006	2088	2206	**2408**	2332	2428	2565	**2800**	2420	2520	2661	**2905**
	kW	*1760*	*1832*	*1935*	*2112*	*1604*	*1670*	*1765*	*1926*	*1865*	*1942*	*2052*	*2240*	*1936*	*2015*	*2130*	*2324*
LSA 52.3 L9	kVA	2360	2445	2583	**2832**	2150	2230	2356	**2572**	2500	2592	2740	**2990**	2600	2688	2840	**3100**
	kW	*1888*	*1956*	*2067*	*2266*	*1720*	*1784*	*1885*	*2057*	*2000*	*2074*	*2191*	*2392*	*2080*	*2150*	*2272*	*2480*
LSA 52.3 L12	kVA	2500	2601	2748	**3000**	2280	2376	2510	**2740**	2650	2757	2913	**3180**	2750	2861	3023	**3300**
	kW	*2000*	*2081*	*2198*	*2400*	*1824*	*1900*	*2008*	*2192*	*2120*	*2206*	*2330*	*2544*	*2200*	*2290*	*2418*	*2640*

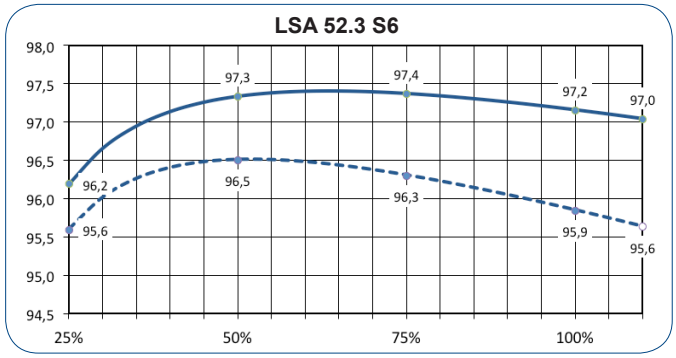
Low Voltage Alternators - 4 pole

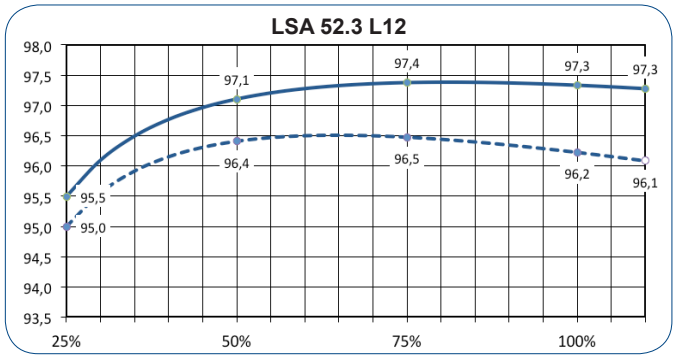
LSA 52.3 - 1860 to 2500 kVA - 50 Hz / 2230 to 3000 kVA - 60 Hz

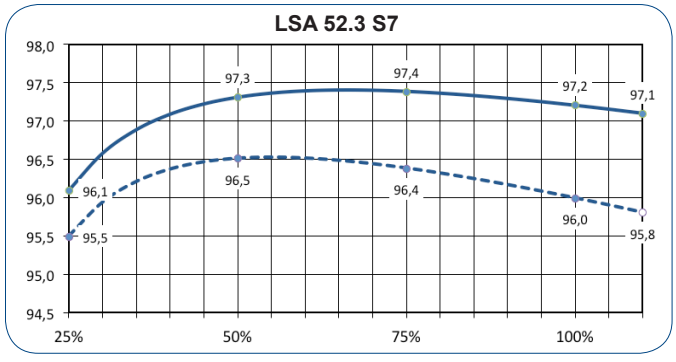
Efficiencies 400V - 50 Hz (— P.F.: 1) (····· P.F.: 0.8)









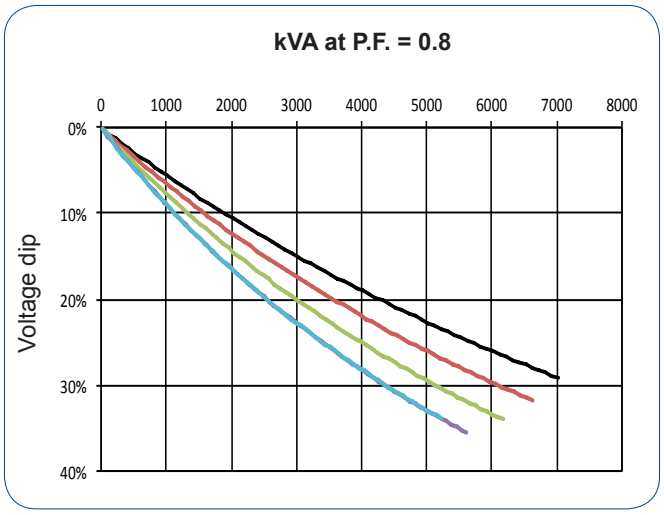


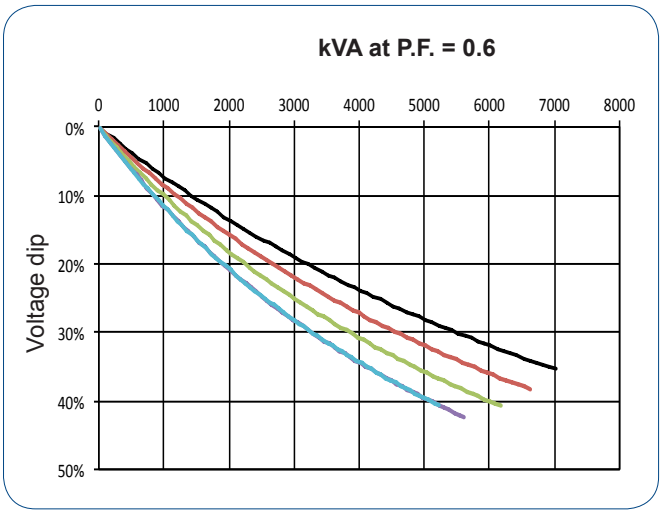
Reactances (%). Time constants (ms) - Class H / 400 V

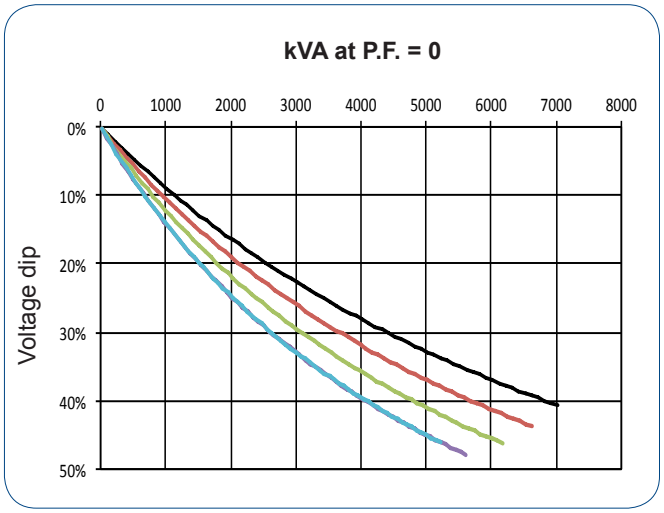
		S5	S6	S7	L9	L12
Kcc	Short-circuit ratio	0.35	0.32	0.35	0.39	0.43
Xd	Direct-axis synchronous reactance unsaturated	367	380	376	344	306
Xq	Quadrature-axis synchronous reactance unsaturated	187	194	192	175	156
T'do	No-load transient time constant	2760	2760	2870	2990	3150
X'd	Direct-axis transient reactance saturated	28.7	30.9	28.9	26.1	23
T'd	Short-circuit transient time constant	254	264	260	267	279
X"d	Direct-axis subtransient reactance saturated	15	16.4	14.8	13.2	11.5
T"d	Subtransient time constant	23	23	22	22	22
X"q	Quadrature-axis subtransient reactance saturated	15.6	16.9	15.4	13.7	12
X0	Zero sequence reactance unsaturated	2.3	2.5	2.6	2.5	2.7
X2	Negative sequence reactance saturated	15.3	16.7	15.1	13.4	11.7
Ta	Armature time constant	28	28	28	28	28
Other class H/400 V data						
io (A)	No-load excitation current	1.1	1.1	1.2	1.2	1.2
ic (A)	On-load excitation current	4.8	5.1	5	4.7	4.3
uc (V)	On-load excitation voltage	55	58	57	53	49
kW	No-load losses	18	18	20	23	25
kW	Heat dissipation	66	74	78	80	82

Transient voltage variation at load inrush: 400V - 50 Hz

L12 L9 S7 S6 S5

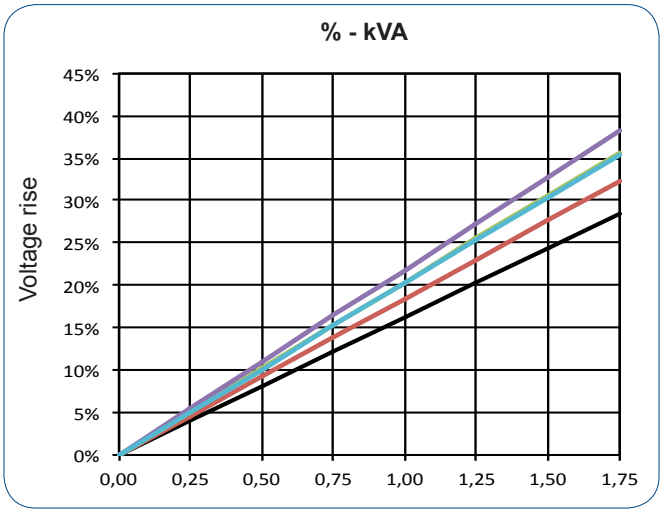






Transient voltage variation at load rejection: 400V - 50 Hz

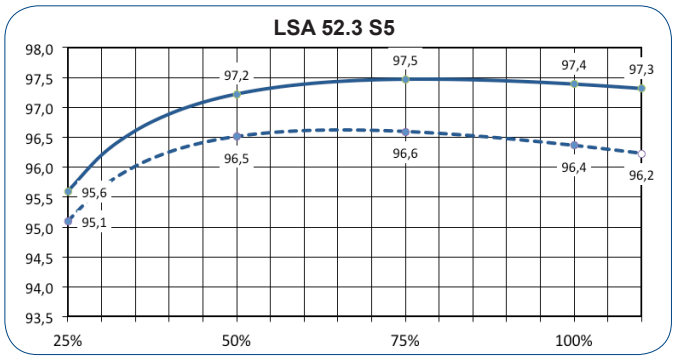
L12 L9 S7 S6 S5

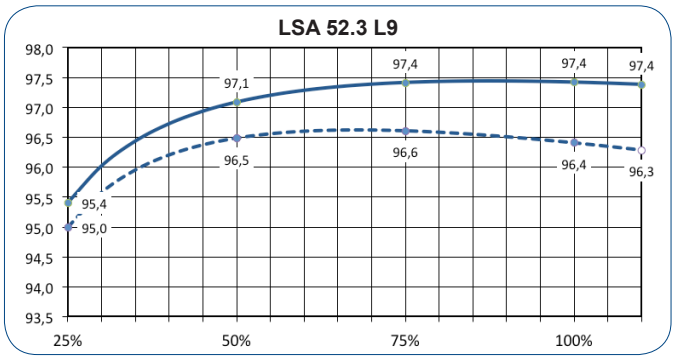


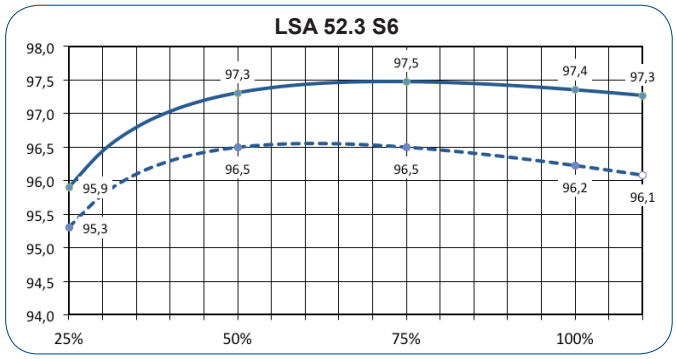
Low Voltage Alternators - 4 pole

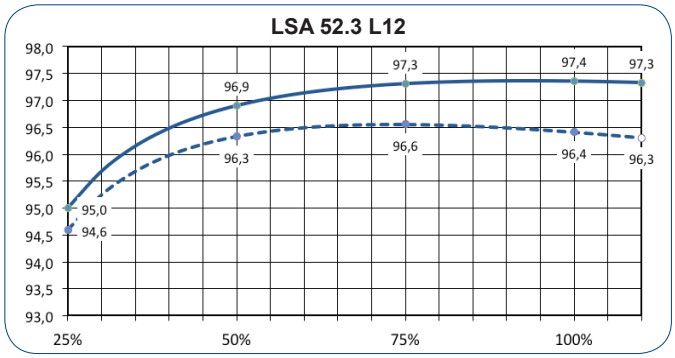
LSA 52.3 - 1860 to 2500 kVA - 50 Hz / 2230 to 3000 kVA - 60 Hz

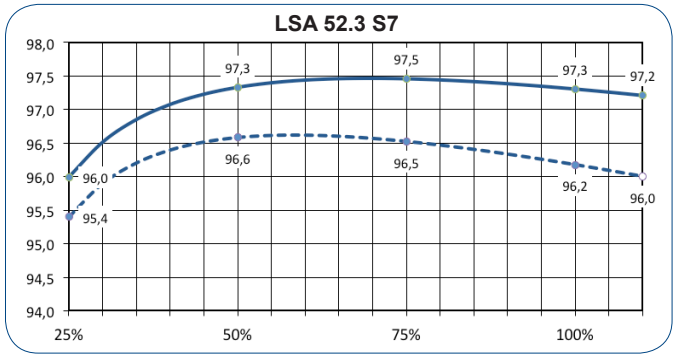
Efficiencies 480V - 60 Hz (— P.F.: 1) (····· P.F.: 0.8)







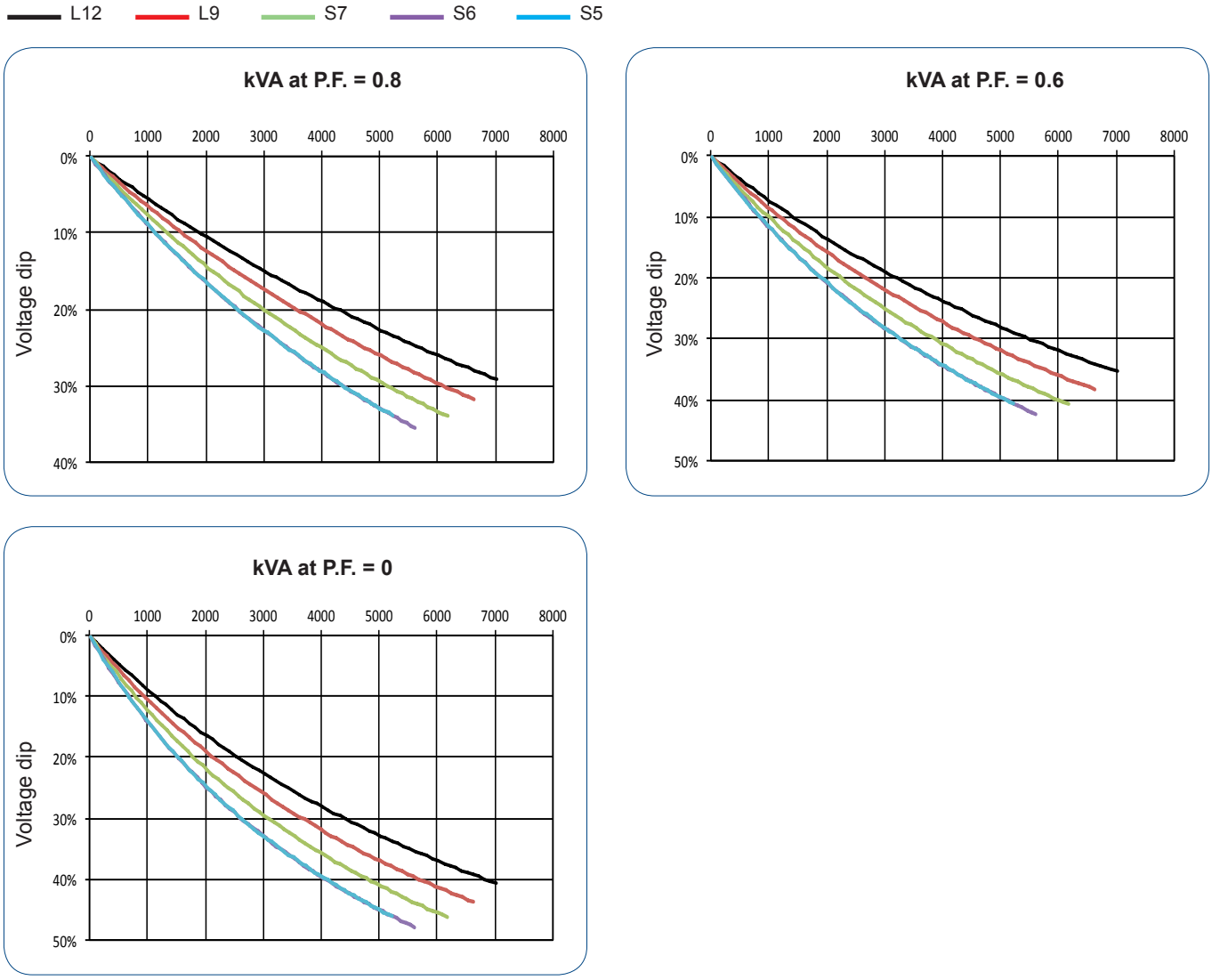




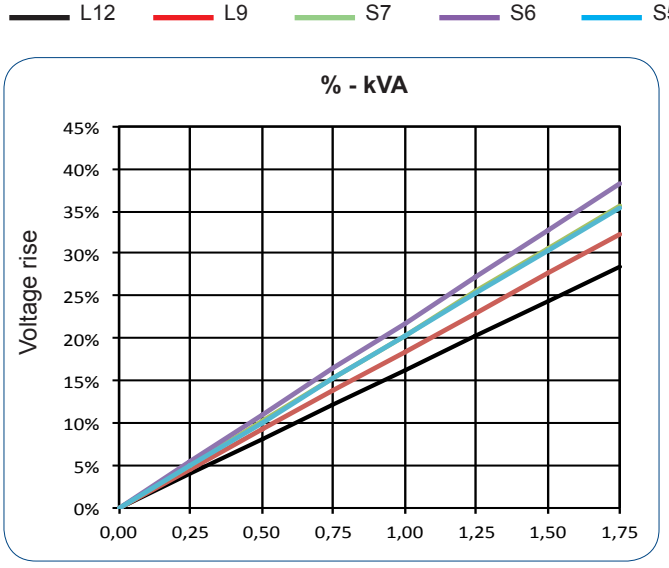
Reactances (%). Time constants (ms) - Class H / 480 V

		S5	S6	S7	L9	L12
Kcc	Short-circuit ratio	0.35	0.32	0.35	0.39	0.43
Xd	Direct-axis synchronous reactance unsaturated	367	380	376	344	306
Xq	Quadrature-axis synchronous reactance unsaturated	187	194	192	175	156
T'do	No-load transient time constant	2760	2760	2870	2990	3150
X'd	Direct-axis transient reactance saturated	28.7	30.9	28.9	26.1	23
T'd	Short-circuit transient time constant	254	265	260	267	279
X"d	Direct-axis subtransient reactance saturated	15	16.4	14.8	13.2	11.5
T"d	Subtransient time constant	23	23	22	22	22
X"q	Quadrature-axis subtransient reactance saturated	15.6	16.9	15.4	13.7	12
X0	Zero sequence reactance unsaturated	2.3	2.6	2.6	2.5	2.7
X2	Negative sequence reactance saturated	15.3	16.7	15.1	13.4	11.7
Ta	Armature time constant	28	28	28	28	28
Other class H/480 V data						
io (A)	No-load excitation current	1.1	1.1	1.2	1.2	1.2
ic (A)	On-load excitation current	4.6	5.1	4.8	4.6	4.2
uc (V)	On-load excitation voltage		58	54	52	48
kW	No-load losses	24	24	27	31	36
kW	Heat dissipation	66	82	85	87	92

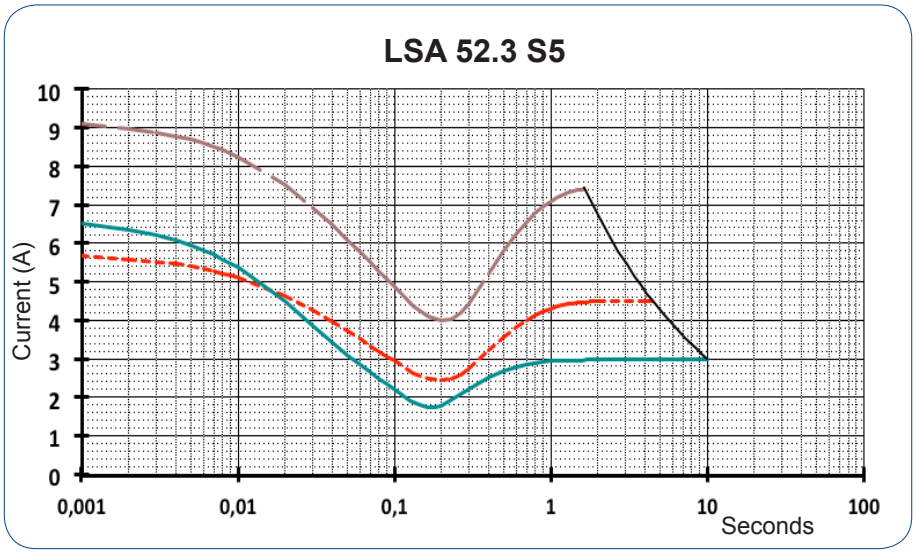
Transient voltage variation at load inrush: 480V - 60 Hz

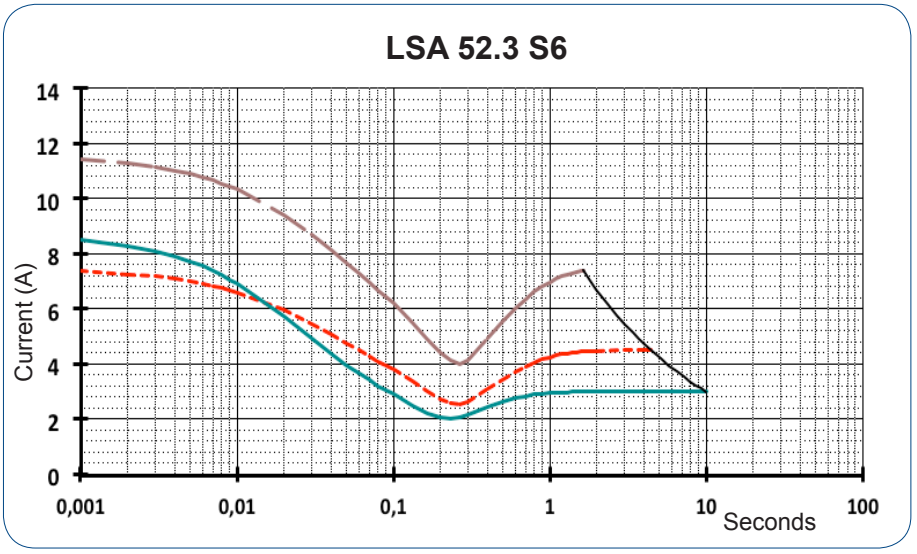


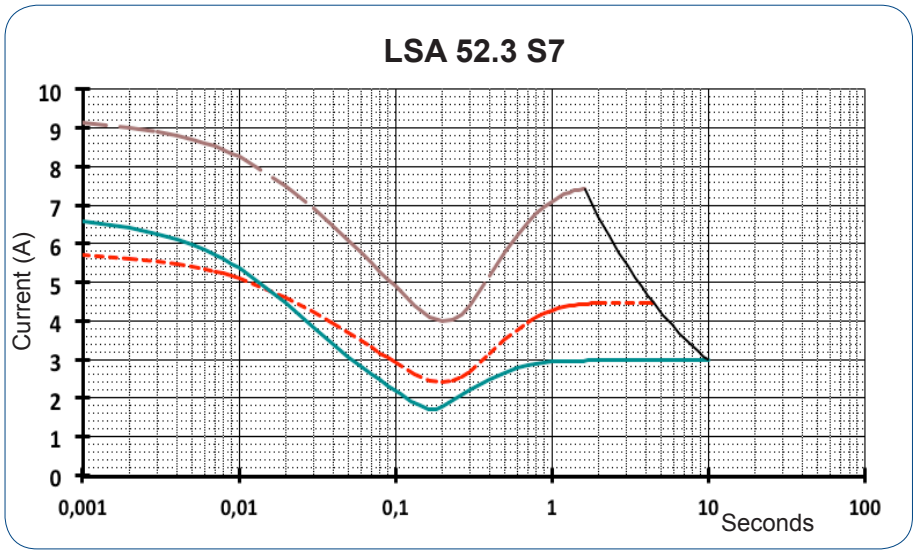
Transient voltage variation at load rejection: 480V - 60 Hz



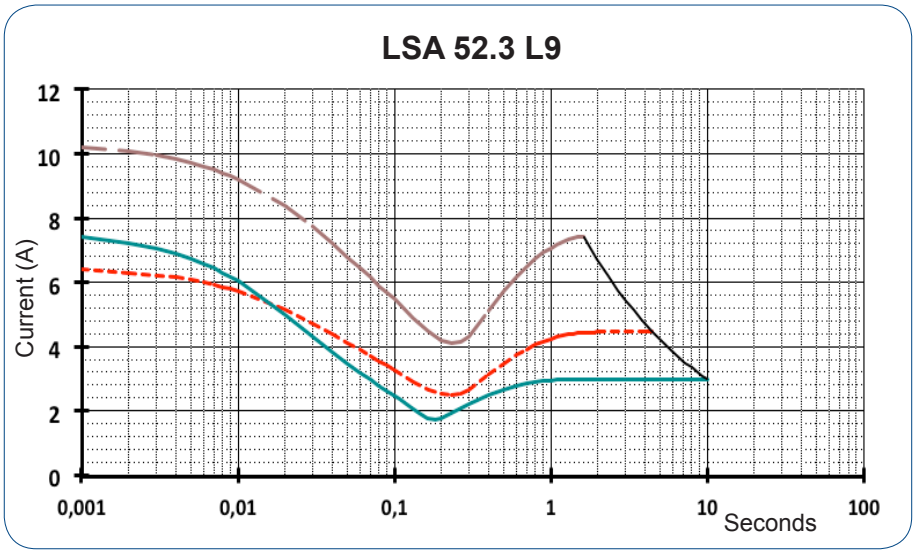
Short-circuit curves

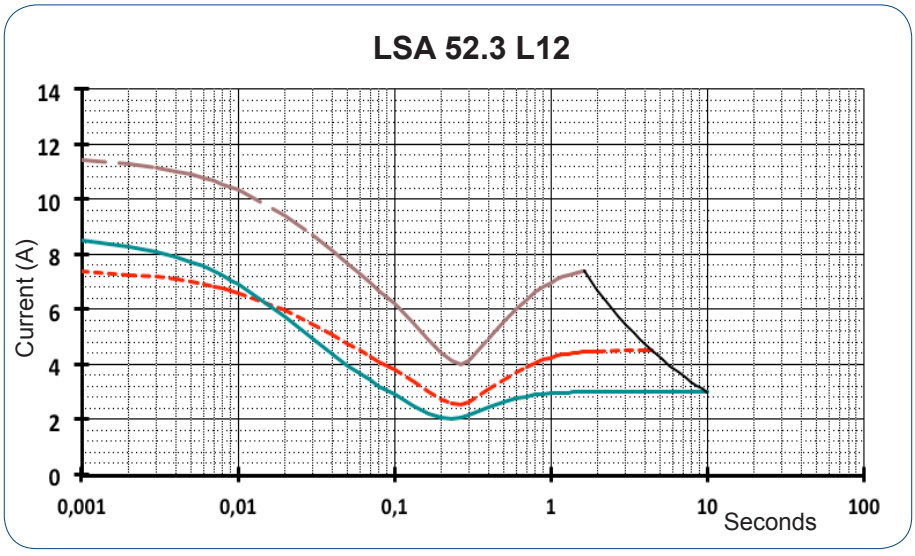






Short-circuit curves



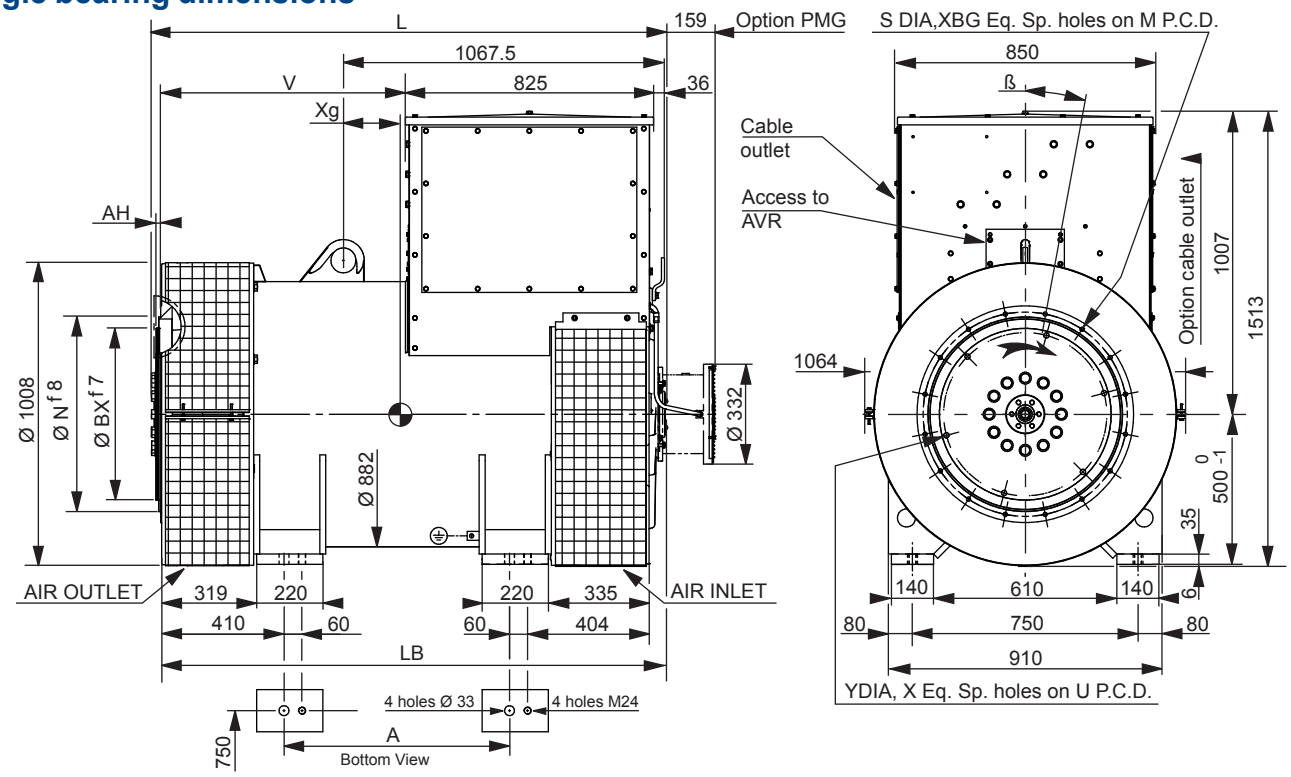


- Symetrical phase to neutral short circuit
- Symetrical two phase short circuit
- Symetrical three phase short circuit
- Heat damage curve limit

Low Voltage Alternators - 4 pole

LSA 52.3 - 1860 to 2500 kVA - 50 Hz / 2230 to 3000 kVA - 60 Hz

Single bearing dimensions



Dimensions (mm) and weight						
Type	L without PMG	LB	A	V	Xg	Weight (*) (kg)
LSA 52.3 S5	1713	1683	750	814	187	3705
LSA 52.3 S6	1713	1683	750	814	187	3705
LSA 52.3 S7	1713	1683	750	814	207	3950
LSA 52.3 L9	1913	1883	950	1014	78	4433
LSA 52.3 L12	1913	1883	950	1014	112	4924

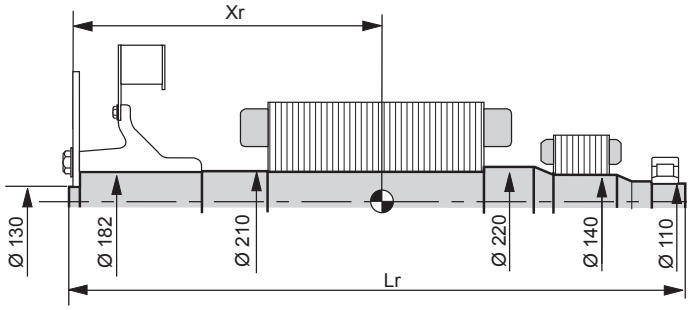
(*) values for S.A.E. 00/21

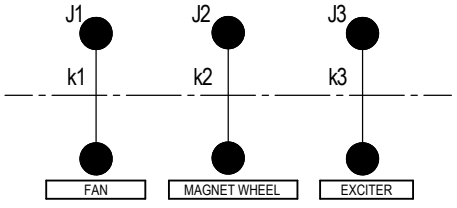
Coupling		
Flange S.A.E.	**0**	**00**
Flex plate S.A.E. **21**		X
Flex plate S.A.E. **18**	X	X

Flange (mm)					
S.A.E.	N	M	XBG	S	β°
0	647.7	679.5	16	14	11°15'
00	787.4	850.9	16	14	11° 15'

Flex plate (mm)					
S.A.E.	BX	U	X	Y	AH
21	673.1	641.3	12	18	0
18	571.5	542.9	6	18	15.8

Torsional analysis data





Centre of gravity: Xr (mm), Rotor length: Lr (mm), Weight: M (kg), Moment of inertia: J (kgm²) : (4J = MD²)								
	Flex plate S.A.E. 18				Flex plate S.A.E. 21			
Type	Xr	Lr	M	J	Xr	Lr	M	J
LSA 52.3 S5	720.4	1689	1420	43.4	702.8	1689	1424	44.3
LSA 52.3 S6	720.4	1689	1420	43.4	702.8	1689	1424	44.3
LSA 52.3 S7	741.5	1689	1453	45	723.9	1689	1457	45.9
LSA 52.3 L9	811.3	1889	1635	50	793.7	1889	1639	50.9
LSA 52.3 L12	858.4	1889	1808	56.5	840.9	1889	1812	57.4

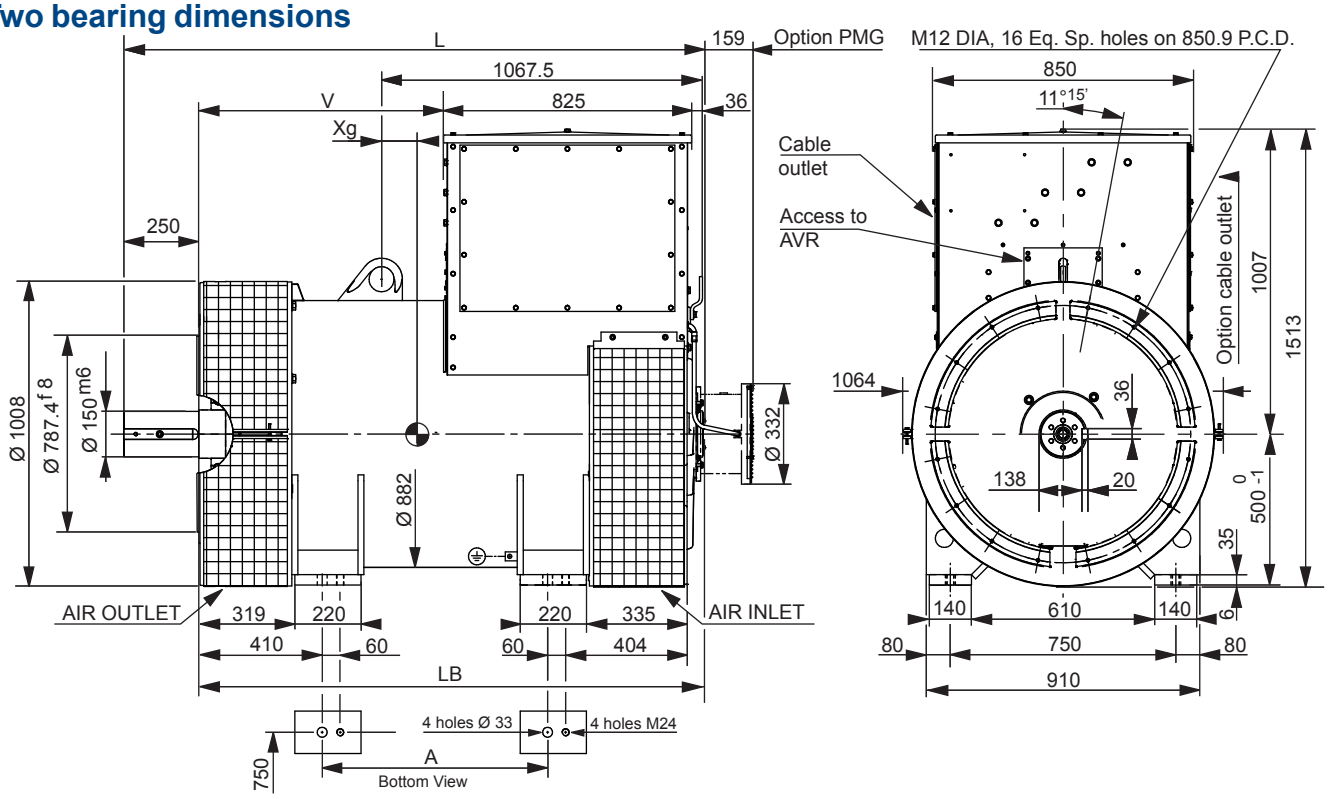
Torsional rigidity					
[Nm/rad]			(kg.m²)		
k1	k2	k3	J1	J2	J3
5.00 10E7	2.50 10E7	1.54 10E7	10.3	32.4	1.5
5.00 10E7	2.50 10E7	1.54 10E7	10.3	32.4	1.5
5.00 10E7	2.41 10E7	1.59 10E7	10.3	34.1	1.5
5.00 10E7	2.29 10E7	1.42 10E7	10.3	39	1.6
5.00 10E7	2.14 10E7	1.52 10E7	10.3	45.6	1.4

NOTE : Dimensions are for information only and may be subject to modifications. Contractuel 2D drawings can be downloaded from the Leroy-Somer site, 3D drawing files are available upon request.

Low Voltage Alternators - 4 pole

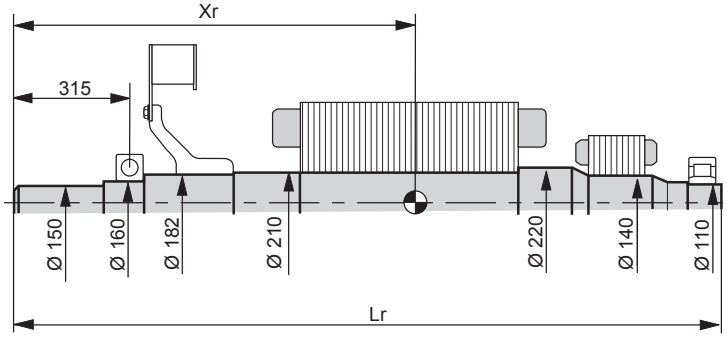
LSA 52.3 - 1860 to 2500 kVA - 50 Hz / 2230 to 3000 kVA - 60 Hz

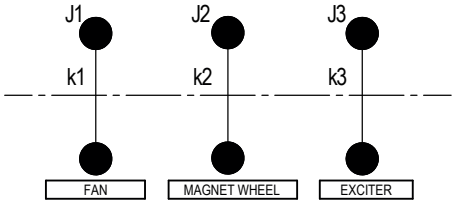
Two bearing dimensions



Dimensions (mm) and weight						
Type	L without PMG	LB	A	V	Xg	Weight (kg)
LSA 52.3 S5	1933	1683	750	814	192	3748
LSA 52.3 S6	1933	1683	750	814	192	3748
LSA 52.3 S7	1933	1683	750	814	212	3991
LSA 52.3 L9	2133	1883	950	1014	83	4476
LSA 52.3 L12	2133	1883	950	1014	117	4967

Torsional analysis data





Centre of gravity: Xr (mm), Rotor length: Lr (mm), Weight: M (kg), Moment of inertia: J (kgm²) : (4J = MD²)				
Type	Xr	Lr	M	J
LSA 52.3 S5	973.4	1912	1363	41.7
LSA 52.3 S6	973.4	1912	1363	41.7
LSA 52.3 S7	994.5	1912	1396	43.3
LSA 52.3 L9	1064.3	2112	1578	48.3
LSA 52.3 L12	1110.6	2112	1752	54.8

Torsional rigidity					
[Nm/rad]			(kg.m²)		
k1	k2	k3	J1	J2	J3
1.82 10E7	3.05 10E7	1.54 10E7	7.7	32.4	1.5
1.82 10E7	3.05 10E7	1.54 10E7	7.7	32.4	1.5
1.82 10E7	2.91 10E7	1.59 10E7	7.7	34.1	1.5
1.82 10E7	2.74 10E7	1.42 10E7	7.7	39	1.6
1.82 10E7	2.53 10E7	1.52 10E7	7.7	45.6	1.4

NOTE : Dimensions are for information only and may be subject to modifications. Contractuel 2D drawings can be downloaded from the Leroy-Somer site, 3D drawing files are available upon request.

EMERSON. CONSIDER IT SOLVED.™

www.emersonindustrial.com

© Emerson 2014. The information contained in this brochure is for guidance only and does not form part of any contract. The accuracy cannot be guaranteed as Emerson have an ongoing process of development and reserve the right to change the specification of their products without notice.

Moteurs Leroy-Somer SAS. Headquarters: Bd Marcellin Leroy, CS 10015, 16915 Angoulême Cedex 9, France. Share Capital: 65 800 512 €, RCS Angoulême 338 567 258.

Leroy-Somer™

EMERSON™
Industrial Automation

5199 en - 2015.04 / b